Youngmin Kim

M.S. Student, Department of Industrial and Systems Engineering
Dongguk University, Seoul, Republic of Korea
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Research Interests

Interpretable Machine Learning · Operations Research · Scheduling · Intelligent Manufacturing

Education

2028 (Expected) — **M.S. in Industrial and Systems Engineering**
Dongguk University, Seoul, Republic of Korea
Advisor: Dr. Sungbum Jun

2025 — **B.S. in Industrial and Systems Engineering**
Dongguk University, Seoul, Republic of Korea
GPA: 3.73/4.00 · Major GPA: 3.86/4.00 · **Graduation Honors**

Selected Major Coursework (A+): Management Science 1, Machine Learning 1, Production & Operations Management, Design and Analysis of Experiment, Advanced Manufacturing Processes, and Logistics

Working Papers

[1] **Kim, Y.**, Lee, C., & Jun, S. (2026). DWARF: An Evolutionary Discretization Framework for Accurate and Interpretable Depth-Limited Decision Trees. *Under revision at Swarm and Evolutionary Computation*.

[2] **Kim, Y.**, Lee, C., & Jun, S. (2026). BONSAI: A Matheuristic for Depth-Constrained Decision Tree Induction with Multi-Way Splits. *Under review at INFORMS Journal on Computing*.

[3] Kang, Y., Kim, S., **Kim, Y.**, Jun, S., & Park, I.-B. (2026). Distributed Flexible Job-Shop Scheduling with Single Minimum-Batch Transfer: Optimization, Heuristics, and Interpretable AI-Assisted Dispatching. *Under revision at International Journal of Production Research*.

Research Experience

Feb.–Aug. 2026 — **Data Inspiring Camp — Manufacturing Data Analysis**
Kia Motors
Decision-tree-based anomaly detection for manufacturing inspection data

Industry Experience

Jul.–Dec. 2024 — **Production Planning Intern**
Continental, Sejong, Republic of Korea

Presentations

International Conferences

[1] **Kim, Y.**, Kim, D., Lee, C., & Jun, S. (2026). Minimizing Total Weighted Tardiness in Parallel Machine Scheduling with Sequence-Dependent Setup Times via Fine-Tuned Large Language Models. *IISE Annual Conference & Expo 2026*, Texas, USA.

[2] **Kim, Y.**, Kim, D., Lee, C., & Jun, S. (2026). Minimizing Total Weighted Tardiness in Parallel Machine Scheduling with Sequence-Dependent Setup Times via Fine-Tuned Large Language Models. *JIMA 2026*, Hakodate, Japan.

[3] **Kim, Y.**, Lee, C., & Jun, S. (2026). BONSAI: A Matheuristic for Depth-Constrained Decision Tree Induction with Multi-Way Splits. *INFORMS Annual Meeting 2026*, San Francisco, CA, USA.

Domestic Conferences

[1] **Kim, Y.**, Lee, C., & Jun, S. (2026). BONSAI: A Matheuristic for Depth-Constrained Decision Tree Induction with Multi-Way Splits. *2026 Spring Conference of the Korean Society of Digital Industry*, Seongnam, Republic of Korea.

[2] **Kim, Y.**, Lee, C., & Jun, S. (2026). BONSAI: A Matheuristic for Depth-Constrained Decision Tree Induction with Multi-Way Splits. *2026 Fall Conference of the Korean Operations Research and Management Science Society (KORMS)*, Seoul, Republic of Korea.

Workshops & Training

Jun. 9–11, 2026 — **SME Week — Smart Manufacturing Training Program**
Apple Manufacturing R&D Accelerator, POSTECH, Pohang, Republic of Korea

Jan. 26–30, 2026 — **2025/26 CALOP Winter Lecture**
Center for Algorithm and Optimization (CALOP), POSTECH, Pohang, Republic of Korea

Technical Skills

Programming: Python · **Optimization:** CPLEX · **Simulation:** AnyLogic · **Typesetting:** LaTeX

Awards & Scholarships

- **Dongguk Founding Scholarship, 2nd Cohort** · Full-Tuition Scholarship · 2023–2025
- **Honorable Mention, Undergraduate Student Project Competition** · Korean Institute of Industrial Engineers (KIIE) · Oct. 2024
  Optimization of Waste Collection Vehicle Scheduling Using Metaheuristic Algorithms